Paleomagnetism and K-Ar Dating of Some Franklin Lavas and Diabases, Victoria Island

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Lavas of the Natkusiak Formation and related diabase sills, Victoria Island, have single polarity remanence directions, which yield a paleomagnetic pole at 163°E, 07°S, which is similar to that obtained from the Coronation Sills of the adjacent mainland. These western Arctic poles are slightly, but significantly different from those obtained from Franklin sills and dikes exposed in the central and eastern Arctic. K–Ar data indicate no resolvable difference in age between the western and eastern Arctic rocks. Although the apparent K–Ar ages from the Victoria Island lavas and sills are discordant, the highest apparent ages are in agreement with a K–Ar isochron age of 625 m.y. obtained from samples of the least altered lavas.

Les laves de la Formation Natkusiak et des sills de diabases associés, sur l'Ile Victoria, ont des directions de polarisation rémanente constantes qui donnent un pôle à 163°E, 07°S, qui est semblable à celui obtenu sur les sills Coronation sur un terrain adjacent. Ces pôles de l'Arctique ouest sont différents de ceux obtenus sur les sills Franklin et les dykes qui affleurent au centre et à l'est du continent Artique. Des dates de K–A n'indiquent aucune différences d'âges entre les roches de l'ouest et celles de l'est. Même si l'âge K–A apparent des laves de l'Ile Victoria et des sills sont discordants, l'âge le plus vieux est en accord avec l'isochrone K–A de 625 Ma obtenu sur les échantillons des laves les moins altérées. [Traduit par le journal]

Introduction

Through the work of Fahrig *et al.* (1971), Robertson and Baragar (1972), and Fahrig and Schwarz (1973), it has been shown that a broad belt of the Canadian Arctic, extending from Baffin Island to the Coronation Gulf, was the site of widespread tholeiitic volcanism in late Precambrian time. Although chemical, petrological, and geochronological data serve to distinguish these rocks from older basaltic rocks in this region (*e.g.*, the Coppermine Lavas and the Mackenzie dikes), the paleomagnetic method has proved to be the most rapid in that these younger basaltic rocks are characterized by paleomagnetic directions, which are usually distinguishable from those of the older Mackenzie dikes and Coppermine Lavas (Park 1974). The younger basaltic rocks are now assigned to the Franklin Magnetic Division on the basis of pole position and by the presence of at least one reversal of magnetic polarity. The older Mackenzie Magnetic Division is characterized by a slightly different pole position and by the absence of polarity reversals. To date, our paleomagnetic knowledge of the rocks comprising the Franklin Magnetic Division is derived from intrusive sills and dikes.

Probable extrusive equivalents of the Franklin sills and dikes are exposed on Victoria Island in the form of the Natkusiak Formation (Thorsteinsson and Tozer 1962), which consists of approximately 1000 ft (330 m) of dominantly basalt flows (Christie 1964). The objectives of the present study are to ascertain, on the basis of paleomagnetic and K–Ar data, whether the Natkusiak flows are indeed correlative with Franklin Magnetic Division rocks. Also the possibility exists that this lava sequence will give information of the reversal history of the Franklin Magnetic Interval. In addition, diabase sills and dikes intrude the Natkusiak Formation and the underlying formations of the Shaler Group of Hadrynian (Late Precambrian) age.

Transportation difficulties in the field made it impossible to sample more than the lower half of the basalt sequence and only a few of the sills. We present the paleomagnetic and geochronologic results so far obtained because to obtain a more complete sampling of these Victoria Island rocks, although desirable, can only be achieved with costly logistical support.

Sampling and Laboratory Methods

(a) Paleomagnetic

A total of 58 independently oriented samples was collected from 18 sites in two separate sub-

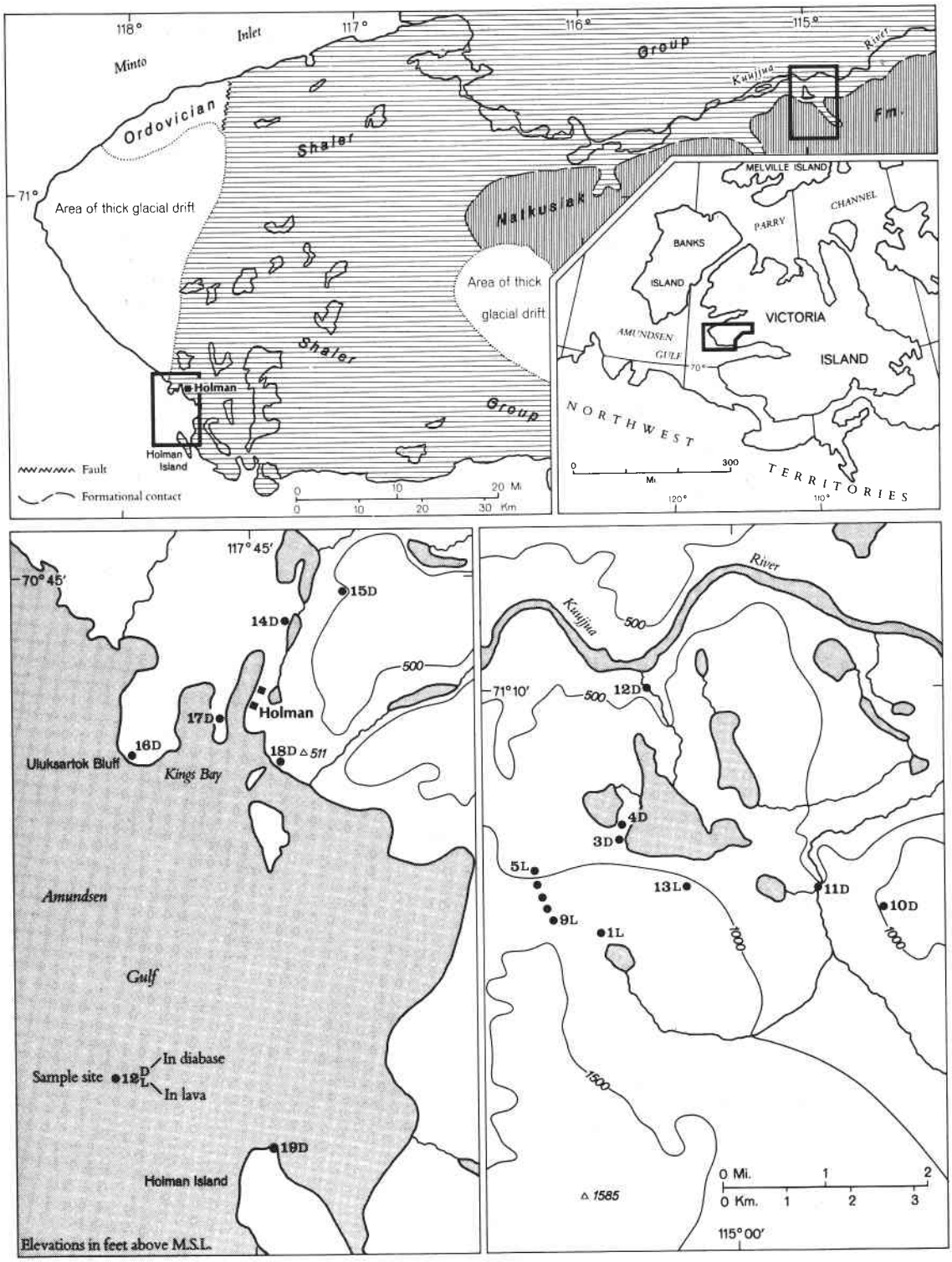


FIG. 1. Location of sampling sites, Victoria Island.

areas of the Minto Arch of Victoria Island (Fig. 1). Sites 1, 5–9, and 13 are within the Natkusiak Formation; the remaining sites are within diabase sheets. The samples are of both the field-drilled and block type, and at all but two sites, the azimuthal component of orientation was by means of a sun compass; at these two sites the azimuthal component of orientation was achieved by sighting to a distant landmark. Two specimens were prepared from each sample and NRM directions and intensities were measured on a commercial spinner magnetometer. The average of the two specimens was used as the basic datum. Following the initial NRM measurements each specimen was partially demagnetized by the alternating field method in steps of 50 Oe (peak) to a maximum of 300 Oe and thereafter in 100 Oe steps to a maximum of 500 Oe (peak). Sample mean directions and site-mean directions were calculated after each stage of demagnetization and the results tabulated. The cleaning field chosen to represent each site (recorded in Table 1) was based upon the achievement of a stable end point; within the range of fields where systematic movement of the site-mean direction was negligible, the maximum within-site precision estimate was used to select the actual cleaning field to represent each site. Samples from two of the sites were also investigated by the thermal demagnetization method. The heatings and coolings were performed in a non-inductively wound furnace enclosed within a commercially available multilayered magnetic shield. The furnace is surrounded by a layer of asbestos insulation, which in turn is surrounded by a water jacket designed to minimize thermal stresses which may affect the properties of the coaxial multilayer shield. To further minimize the effects of uncancelled magnetic fields, the sample container is slowly rotated during the heating and cooling cycles.

At least one thin section and polished section was prepared from each site and preliminary examination of these thin sections, governed to some extent, our choice of samples for potassium and argon analyses.

(b) K–Ar Dating

Samples were crushed to pass through a 20 mesh screen. Powders finer than 100 mesh were discarded. The sample was then homogenized and split into four aliquots. One aliquot was further crushed to pass 100 mesh and used for potassium analysis. Potassium analysis was performed with an Instrumentation Laboratory flame photometer Model 143, with a lithium standard and sodium buffer. Argon was measured by the isotope dilution method with an AEI MS-10 mass spectrometer.

For the K–Ar isochron study, a crucible made of 0.01-in. (0.25 mm) thick molybdenum sheet was degassed in an argon extraction system, and then the system was opened to the atmosphere. The sample, which had been crushed within 24 hours, was loaded and the system was evacuated for about 28 hours, including 18 hours of baking at 170°C. The blank ^{40}Ar and ^{36}Ar, which are considered to be the amount of contamination added during the argon analysis, were subtracted from the results in the calculations of the initial ^{36}Ar content and $^{40}Ar/^{36}Ar$ and $^{40}K/^{36}Ar$ ratios. The blank ^{36}Ar corrections amounted to 36% for N-20 and 16% for N-21, but the reproducibility of blank runs assured the reliability of the correction.

For the conventional K–Ar dating, most of the procedures necessary for minimizing the genuine atmospheric argon contamination were not applied. A sample was loaded in an undegassed crucible and pumped overnight with or without a short baking.

Paleomagnetic Results

The paleomagnetic directional data are graphically displayed in Fig. 2 and are tabulated in Table 1. The initial directions display a streaked pattern between the present field direction and a presumed older direction having an east-northeast declination and shallow inclination. Alternating field demagnetization removes most of this streaking and the directions converge around a mean of 79°, + 7°. Two lava sites however, 1L and 9L, retain high to moderately high values of inclination throughout all stages of AF treatment in spite of the fact that the intensities of all samples constituting these sites are reduced in intensity to 1 or 2% of their initial values. These sites are in two separate flows at the highest stratigraphic levels sampled (400 ft to 500 ft (122 to 152 m) above the base of the lava sequence). They may be recording a genuine field excursion in the Franklin Magnetic Interval, or they may have been remagnetized. If they have been remagnetized, the AF method is unsuccessful in erasing secondary components

TABLE 1. Paleomagnetic directional data

		NRM				'Cleaned'					Pole position	
Site	*N*	*D*	*I*	*R*	*AF*	*D*	*I*	*R*	*K*	α95	Long.	Lat.
1L	2	055.3	+83.4	1.994	500	058.2	+76.6	1.996	262	15.5°	143.2°E	68.1°S
3D	3	132.8	−54.3	2.984	200	082.7	+09.9	2.997	677	4.7°	160.3°E	7.1°S
4D	3	253.7	+80.0	2.982	500	071.2	+10.1	2.968	62.4	15.7°	171.3°E	10.8°S
5L	3	073.0	+32.0	2.485	500	080.0	+00.4	2.802	10.1	41.0°	164.4°E	3.4°S
6L	3	076.3	+58.7	2.389	150	079.1	+12.3	2.827	11.6	38.0°	163.3°E	9.4°S
7L	3	069.6	+39.5	2.776	500	066.3	−02.7	2.647	5.7	57.6°	178.0°E	6.2°S
8L	2	084.4	+83.2	1.997	300	093.4	+24.6	1.995	198	17.8°	147.6°E	11.1°S
9L	3	079.9	+59.4	2.909	250	052.2	+51.3	2.822	11.3	38.6°	180.6°E	42.0°S
10D	3	027.8	+49.2	2.146	500	070.6	+12.4	2.925	26.6	24.4°	171.5°E	12.1°S
11D	3	069.3	+67.7	2.937	500	078.2	+13.1	2.973	73.3	14.5°	164.1°E	10.1°S
12D	3	061.6	+71.9	2.992	400	072.4	+17.8	2.978	92.7	12.9°	168.8°E	14.3°S
13L	3	123.9	+87.0	2.983	150	087.0	+18.3	2.892	18.4	29.6°	154.8°E	9.9°S
14D	3	066.4	+44.3	2.893	400	083.3	+00.6	2.985	136	10.6°	158.5°E	2.5°S
15D	3	078.5	+43.0	2.947	250	078.8	+05.6	2.992	242	7.9°	162.0°E	6.3°S
16D	4	066.8	+23.4	3.754	200	078.0	−11.6	3.972	106	9.0°	165.5°E	1.6°N
17D	7	078.9	+35.3	6.470	300	082.4	+03.1	6.873	47.2	8.9°	159.0°E	4.0°S
18D	3	083.0	+35.2	2.872	150	086.7	+00.2	2.983	118	11.4°	155.4°E	1.2°S
19D	4	074.7	+12.7	3.984	300	076.1	−03.8	3.986	217	6.3°	166.1°E	2.8°S
Mean	16		(South) Pole Position (1)						89	4.0°	163°E	7°S

NOTES: *N* = number of samples collected at the site. *D* = Declination, *I* = Inclination. *AF* = Cleaning Field in Oersteds. *R*, *K*, α95 are the standard statistical parameters of Fisher (1953).
(1) Mean Pole calculated from site poles with 1L and 9L excluded.

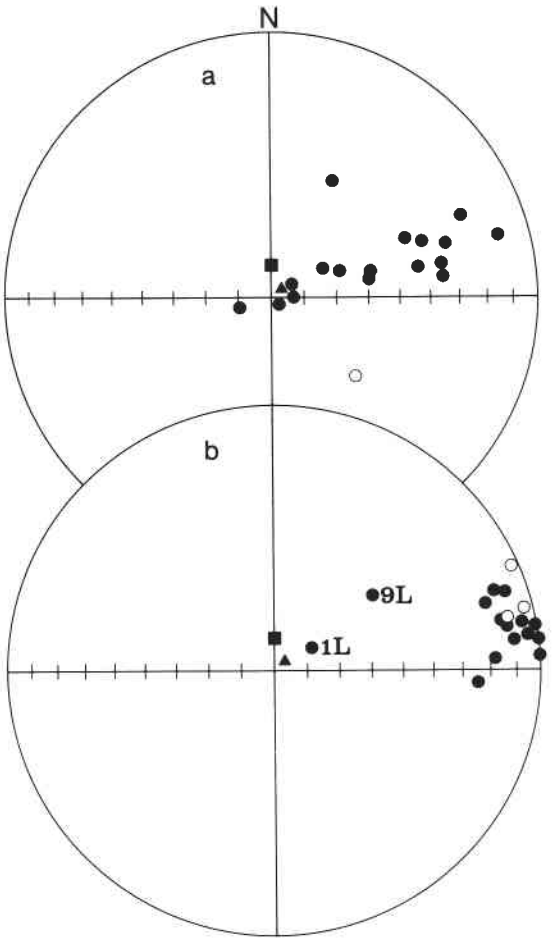


FIG. 2. Site mean NRM directions of 18 sites before a.f. cleaning (a) and after optimum cleaning (b). Solid (open) circles are on the lower (upper) hemisphere. Solid squares and solid triangle are the axial dipole field direction and the present field direction respectively.

and in isolating a primary component if any primary component remains in these samples. Thermal demagnetization was performed on specimens from these two sites subsequent to the alternating field treatment. Although the intensities of magnetization were further reduced by this treatment, the specimens acquired large temporary components of magnetization, presumably in the short time interval between removal from the shielding and placement into the measuring magnetometer. Although the magnetometer readings stabilized after several minutes within the field-free environment of the magnetometer, directional agreement among specimens within a site deteriorated such that the mean within-site direction became statistically insignificant. Because of the uncertain significance of the directional data from sites 1L and 9L, they have been omitted from the calculation of the mean pole position recorded in Table 1. The case for these sites recording a genuine field excursion would be strengthened if subsequent vertical sampling along another laterally separated traverse yielded these directions at the same stratigraphic level.

Petrography and K–Ar Results

(a) Diabases

The diabase samples examined are of fine-to-medium grain size and consist of subequal amounts of plagioclase and clinopyroxene usually displaying an ophitic or sub-ophitic texture. Twelve of the 14 slides examined contain small amounts of olivine or pseudomorphs of olivine. The secondary minerals replacing olivine include pale green and yellow-green serpentine and a colorless, micaceous mineral of high birefringence, which is probably talc. Primary Fe-Ti oxides, in addition to small amounts of magnetite accompanying the olivine alteration, constitute about 5% of the rocks. Diabases without olivine pseudomorphs contain trace amounts of quartz and myrmekite. All slides contain variable amounts of chlorite and sericitic alteration of the plagioclase.

Potassium content (recorded in Table 2) not unexpectedly, correlates positively with the degree of sericitic alteration of the plagioclase. The apparent ages range from 470 m.y. to 635 m.y. As seen in thin section, there is no correlation between apparent age and degree of alteration.

(b) Lavas

The lavas are dark gray to greenish gray rocks of fine grain size. In thin section they display a sub-ophitic to intersertal texture. The least altered lavas consist of subequal amounts of plagioclase, clinopyroxene, and an interstitial chlorite. With increasing alteration calcite accompanies the interstitial chlorite. Amygdaloidal lavas tend to be even more altered, with partial to complete destruction of the clinopyroxenes and their replacement by chlorite, calcite, and sphene. Amygdular fillings include chlorite (generally of a paler color than the groundmass chlorite), calcite, quartz, and chalcedony. In some of the more altered lavas,

TABLE 2. Magnetic properties and K–Ar data

Site No.	n	Jo emu/g × 10^{-5}	M.D.F. NRM	$Q = \frac{Jo}{xH}$	Sample No.	$K\%$	40 Arad × 10^{-6}cm^3/g	Calculated age (m.y.)
1L	4	107 ± 45	87 ± 10	1.1 ± 0.4	N-1	0.200	2.71	310
					N-2	0.540	10.10	415
3D	5	172 ± 49	38 ± 4	3.0 ± 1.4				
4D	6	170 ± 61	65 ± 5	3.1 ± 1.1				
5L	6	34.1 ± 30.1	306 ± 207	0.7 ± 0.4	N-9	1.16	34.66	630
					N-10	1.48	22.96	350
					N-11	2.23	32.67	335
6L	6	33.3 ± 24.0	156 ± 182	0.7±0.2	N-12	0.152	3.77	535
					N-13	1.26	31.86	545
					N-14	0.323	6.34	435
7L	6	26.5 ± 10.2	179 ± 12	0.6 ± 0.3	N-15	1.55	39.50	550
					N-16	1.01	22.73	490
					N-17	0.965	22.15	500
8L	4	33.2 ± 4.2	42 ± 3	0.5 ± 0.1	N-18	1.36	35.43	560
					N-19	1.17	26.60	495
9L	6	73.2 ± 17.6	46 ± 11	1.1 ± 0.2	N-20	0.0946	2.84	630
					N-21	0.0594	1.67	595
					N-22	0.0640	1.61	540
10D	6	560 ± 776	50 ± 10	6.6 ± 8.0				
11D	6	103 ± 32	88 ± 20	0.8 ± 0.2	N-26	1.05	31.89	635
					N-28	0.393	8.62	480
12D	6	47.7 ± 2.6	100 ± 7	1.0 ± 0.1				
13L	6	47.2 ± 8.2	59 ± 12	0.6 ± 0.1	N-32	0.183	4.90	570
					N-33	0.205	5.35	560
					N-34	0.238	6.23	560
14D	6	24.0 ± 3.8	76 ± 6	0.6 ± 0.1	N-36	1.26	33.33	565
					N-37	1.34	37.47	595
15D	6	30.2 ± 7.6	130 ± 27	0.9 ± 0.2	N-38	0.342	7.84	500
					N-39	0.439	11.14	545
16D	8	35.7 ± 11.8	150 ± 19	0.8 ± 0.2	N-41	0.585	14.24	525
					N-42	0.403	8.62	470
					N-43	0.651	14.63	490
					N-44	0.632	15.30	525
17D	10	55.4 ± 33.1	117 ± 32	1.1 ± 0.5	N-45	0.500	11.52	500
					N-47	0.214	4.85	495
18D	6	43.4 ± 9.0	137 ± 23	1.0 ± 0.2	N-48	0.564	12.94	500
					N-49	1.21	26.31	475
19D	4	64.2 ± 3.1	116 ± 28	1.1 ± 0.2				

NOTES: n = the number of specimens used in calculating the means and standard deviations of the means of the remanent magnetization intensity, median destructive alternating field of the NRM, and Koenigsberger ratio for each site.

pyrite is moderately abundant, with a concomitant reduction in the amount of Fe-Ti oxides, which in the least altered lavas constitute 3–4% by volume.

Although some of the plagioclases have sericitic alteration, there is much less visible correlation between sericite and potassium content—a correlation that is positive in the diabases. The apparent ages of the basalts exhibit an even greater spread (310 m.y.–630 m.y.) than do the diabases. There appears to be little or no correlation between apparent age and petrographic alteration.

Lava sites 9 and 13 are in massive flows free of amygdules and are the least altered mineralogically. The six samples from these two sites were chosen for the isochron study; the techniques involved in this study were discussed earlier. The analytical results are given in Table 3 and are graphically displayed in Fig. 3. These samples define a 625 m.y. isochron with an initial argon ratio approaching zero. This isochron age is in agreement with the highest apparent individual ages of both the basalts and diabases. The isochron age is also in good agreement with the mean age calculated by Fahrig

TABLE 3. K–Ar isochron data

Sample No.	K%	^{40}Ar total 10^{-6}cm^3/g	^{36}Ar 10^{-10}cm^3/g	^{40}K/^{36}Ar $\times 10^5$	^{40}Ar/^{36}Ar $\times 10^4$
N-20	0.0946	2.87	1.10	5.89	2.61
N-21	0.0594	1.72	1.56	2.60	1.10
N-22	0.0640	2.03	14.2	0.31	0.14
N-32	0.183	5.20	10.2	1.22	0.51
N-33	0.205	5.65	10.0	1.40	0.56
N-34	0.238	6.71	16.7	1.00	0.41

TABLE 4. Franklin poles from sub regions A and B (Fig. 4)

Sub region	Formation description	Mean pole (N, K, A_{95})
(A) West of 114°W North of 67½°N	Coronation Sills, Victoria Island sills and lavas	*161.1°E, 4.1°S (31, 39, 4.2°)
(B) (See Fig. 4)	Bathurst, Boothia, and Baffin Island dikes and sills	†165.5°E, 8.0°N (34, 42, 3.8°)

NOTES: N is the number of site poles used in the statistical analysis, K is the precision, and A_{95} is the radius of the 95% circle of confidence.

*This pole includes the 13 site poles of Robertson and Baragar (1972), site poles 1 and 2 (Coronation sheet sites) of Fahrig *et al.* (1971), and the 16 site poles of this study. Site poles of this group are plotted as solid dots in Fig. 4.

†This pole includes the remaining 21 site poles of Fahrig *et al.* (1971) (excluding the Group B poles of their Table 2), 10 site poles of Fahrig and Schwarz (1973), and site poles 31, 33, and 16 of Park (1974). Site poles of this group are plotted as open circles in Fig. 4.

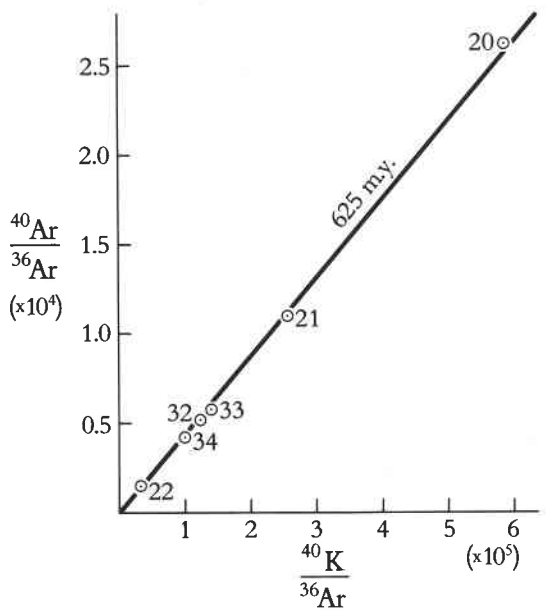


FIG 3. ^{40}Ar/^{36}Ar *versus* ^{40}K/^{36}Ar plot of whole rock data from lava sites 9 and 13.

et al. (1971) from a large number of K–Ar determinations by the Geological Survey of Canada since 1963 on other Franklin rocks.

Fe–Ti Oxide Petrography and Magnetic Properties

(a) Diabases

The major opaque phases consist of separate ilmenite grains, separate titanomagnetite grains, and composite titanomagnetite-ilmenite grains. Minor opaque phases are pyrite, chalcopyrite, and bornite. Based on the density and volume of ilmenite lamellae within the titanomagnetites, most fall in the oxidation classes 2 and 3 of Ade-Hall *et al.* (1968). Further effective reduction of the magnetite grain size is brought about by magnetite-ulvospinel intergrowths as seen in a few of the sections examined. Presumed effects of subsequent burial metamorphism (Ade-Hall *et al.* 1971) are slight. Granulation of magnetite and whitening of the magnetite to titanohematite was observed in about half the sections examined.

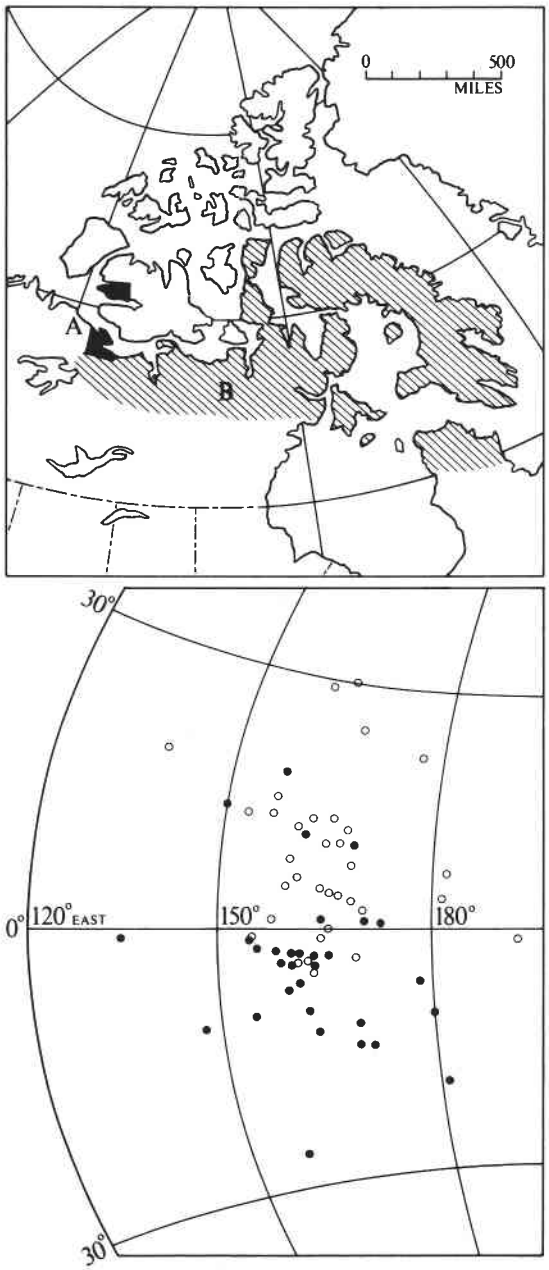


FIG. 4. Solid dots are pole positions from Franklin rocks of sub-region A. Open circles are poles from Franklin rocks from sub-region B.

In common with most diabases, the magnetic mineralogy of these samples displays a limited range of textures and phases present. The simple magnetic properties of median destructive field of NRM and Konigsberger ratio (recorded in Table 2) similarly show a limited range, characteristic of samples of low to moderate magnetic stability. However, the magnetic mineralogy does not, in any simple quantitative way, correlate with those sites exhibiting magnetic properties at either the high or low end of these ranges.

(b) Lavas

In terms of the high temperature oxidation scale of Ade-Hall *et al.* (1968) no grains in these lavas exceed oxidation class 2. However low temperature, possibly burial-induced petrographic alterations are more widespread and variable both on a within-flow and between flow basis. These alterations include: partial replacement of the titanomagnetites by titanohematite; granulation of the titanomagnetites by ferrirutile, in original class 2 grains by the replacement of the ilmenite lamellae by sphene; and by the development of sphene along irregular fractures in class 1 grains (no ilmenite present). In addition, in parts of two lavas, considerable pyrite is present with a concomitant reduction in Fe-Ti oxide content. These within-flow variations in alteration phenomena are reflected in the large spread in both NRM intensity and median destructive field of NRM recorded in Table 2 for lava sites 5 and 6. Lava sites 1 and 9, which yield anomalous directions of remanence, are otherwise similar to other lava sites in terms of opaque petrography and magnetic properties. If these sites are dominated by a secondary component of magnetization, this secondary magnetization is not tied to a unique observable petrological alteration.

Franklin Poles and Discussion

If the 16 paleomagnetic poles calculated from the results of this study are added to the 49 paleomagnetic poles recorded (or calculated from the tabulated and figured data) in the previous publications concerning the Franklin Diabases (Fahrig *et al.* 1971; Robertson and Baragar 1972; Fahrig and Schwarz 1973; Park 1974) a mean pole of 164°E, 2°N, $\alpha 95 = 3.1°$ is obtained.

However this averaging process tends to obscure a feature of the data that may be of some regional significance in stratigraphic correlation studies. When all 65 poles are plotted, a non-Fisherian distribution is apparent; the scatter of poles is much greater in a latitudinal sense than in a longitudinal sense. Poles derived from rocks

on Victoria Island and the Coronation Gulf region west of the 114° meridian and north of the 67½° parallel of latitude tend to plot to the south of those from the Bathurst, Boothia, and Baffin Island sills and dikes. Furthermore sites from the latter region yield poles of both polarities whereas all sites from the Coronation Gulf and Victoria Island region are of one polarity (south poles as plotted in Fig. 4). When the data are so divided, two clusters of poles are produced, with relatively little overlap. The Boothia, Bathurst, and Baffin Island poles accurately reflect the field orientation with respect to this region at the time of initial cooling of the sills and dikes, since the reversals of polarity are exact (see Fahrig and Schwarz 1973, Fig. 2). If the magnetization in the Victoria Island and Coronation Gulf region is also a TRM in origin, the data suggest that there is a real age difference in the timing of igneous activity in these two regions. This time difference may be very small: there is no resolvable difference in age apparent in the K–Ar whole-rock data.

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